HUMAIRA ANZUM NEHA

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Technical Summary

- Research experience in human-centered AI and accessibility for people with visual impairments and people who stutter.
- Research experience in qualitative HCI studies, including workshops, semi-structured interviews, and thematic analysis.
- Research experience in computer vision for object detection, currency recognition, and face recognition.
- Research experience in building assistive wearables with ToF and accelerometer sensors on Raspberry Pi.
- Research experience in developing a mobile app for non-invasive glucose and hemoglobin estimation using PPG signals.
- Hands-on experience in deploying deep learning models on edge devices for real-time use.
- Hands-on experience in collecting and annotating custom image datasets for model training.
- Hands-on experience in deep learning frameworks and libraries e.g., TensorFlow, Keras, PyTorch, Hugging Face.
- Developed a model agnostic training-free method for long video understanding with vision-language models.
- Hands-on experience in graph neural networks and multimodal audio–text fusion with BERT.

Education

M.Sc in CSE at BRAC University (CGPA:4.00/4.00) **Jan. 2026 – Present**
Completed Credit: 12 *Dhaka, Bangladesh*

B.Sc. in CSE at KUET (CGPA:3.87/4.00, 3rd Position) **Feb. 2019 – Mar. 2024**
Thesis: Automated Object Identification for the Visually Impaired *Khulna, Bangladesh*

Skills

Programming Languages: Python, C, C++, Java, Kotlin, Javascript, Bash, and SQL.
Deep Learning Libraries: TensorFlow, Keras, and PyTorch, Hugging Face
AI Tools: OpenAI Codex and Claude.
Developer Tools: Anaconda, Jupyter, Arduino, Raspberry Pi , Git, MS Visual C++, and VS Code.
Mobile Apps Development: Android Studio.
Operating System: Linux, MacOs, and Windows.
Academic Expertise: Teaching, Undergraduate Thesis Supervision, Teaching Assistant Coordination, Competition Mentoring, and Project Evaluation and Judging.
Research Aptitude: Human-Centered AI for Accessibility, Assistive Computer Vision, Qualitative HCI Research, and Multimodal AI with Vision-Language Models.
Adaptive Skills: Mentoring and Leadership, Collaborative Research, Community-Engaged Research, Academic Writing, Complex Problem Solving, and Rapid Skill Acquisition.

Experience

Lecturer **Aug 2024 – Present**
Computer Science and Engineering, United International University *Dhaka, Bangladesh*

- Leading a project studying the reliability of vision-language models in reading and reasoning across diverse expiry date formats for blind and low-vision users, along with strategies to improve their performance.
- Taught undergraduate courses including Data Structures and Algorithms, Database Management Systems, Data Mining, Object Oriented Programming, and Computer Architecture.
- Supervised undergraduate thesis projects, mentored student teams for competitions, and judged departmental project shows.

Projects

AI-Assisted Speech Editing for People Who Stutter | *Qualitative HCI Study* **2026**

- Investigated how people who stutter in Bangladesh use video to represent themselves and their community, their current editing practices, and their expectations of AI-assisted video and speech editing tools.
- Conducted an onboarding workshop and semi-structured interviews with eight participants, analyzed using reflexive thematic analysis, and derived design implications for more inclusive AI tools. Manuscript under review.

Query-Guided Visual Prompting for Training-Free Long Video Understanding | *Python, PyTorch, VLMs* **2026**

- Developed a model-agnostic, query-guided asymmetric grid composition method that arranges selected frames into visual prompts, with no fine-tuning or model changes.
- Using CLIP-based frame scoring with Qwen2.5-VL-7B, improved pair accuracy on the MF2 benchmark from 16.9% to 23.5% over the Video Panels baseline.

Multimodal GNN–BERT Fusion for Music Tagging and Retrieval | *PyTorch, PyTorch Geometric, BERT* **2026**

- Built audio graph models (GraphSAGE, GAT) and fused them with BERT text embeddings for multi-label tagging on MusicCaps, GTZAN, and MagnaTagATune.
- Cross-attention fusion raised micro-F1 on MagnaTagATune from 0.38 (BERT-only) to 0.52.
- CustomiZed MusicCaps-Curated-Tags, a 216-tag vocabulary across 5 categories built with an LLM-assisted pipeline.

Edge-Deployed Object and Currency Recognition for the Visually Impaired | *PyTorch, YOLOv5, ResNet-50* **2024**

- A real-time system to help visually impaired people with object recognition, monetary transactions, and social interactions.
- Built a custom annotated dataset of ∼21,000 images of common objects and Bangladeshi banknotes (object dataset, currency dataset), trained YOLOv5 for object and banknote detection and denomination classification, used ResNet-50 for face recognition, and deployed on a Raspberry Pi.
- Achieved 98.2% mAP for objects, 99.7% top-1 accuracy for currency, and 97.6% accuracy for face recognition.

Smart Assistive Glasses for Navigation, Fall Detection, and Reading | *Raspberry Pi, Tesseract OCR* **2024**

- Developed an assistive smart glass with real-time obstacle avoidance using Time-of-Flight sensors, fall detection with caregiver alerts using an accelerometer, and OCR with text-to-speech for reading printed text.
- Achieved 93.03% character accuracy (6.97% CER) and 9.46% WER for OCR on 50 test documents.

Mobile Application to Measure Blood Component Levels Non-Invasively | *Kotlin, Python, OpenCV* **2023**

- Developed a mobile app for non-invasive hemoglobin and glucose monitoring, designed for users with limited smartphone experience.
- Extracted PPG signals, integrated a deep learning model to estimate blood component levels, and calibrated camera settings for consistent video quality across lighting and devices.
- Achieved $R^2 = 0.81$ and MAE of 0.68 g/dL for hemoglobin against invasive tests

Abalone Board Game with a Minimax AI Opponent | *Python, Pygame* **2023**

- Developed a playable Abalone board game with an AI opponent using Minimax search and alpha–beta pruning.

Smart&Safe: Verified Marketplace for Used Smartphones | *ASP.NET, JavaScript, Bootstrap* **2022**

- Developed a web app for trading used smartphones with NID-based identity verification and automated IMEI screening.

Together: Team Task and Deadline Management App | *Java, Firebase, Android Studio* **2021**

- Developed a mobile app for teams to assign tasks, track responsibilities, and manage shared deadlines.

Relevant Coursework

- Advanced Natural Language Processing
- Deep Learning for Computer Vision
- Neural Networks and Fuzzy Systems
- Research Methodology and Seminar (HCI-Focused)
- Fairness, Ethics and Explainability in AI
- Data Mining
- Artificial Intelligence
- Natural Language Processing
- Image Processing
- Applied Statistics

Publications

- **H. Neha**, M. S. Sadi, M. R. Islam and S. Sakib, "Automated Object Identification for the Visually Impaired," 2024 27th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2024, pp. 3584-3589, doi: 10.1109/ICCIT64611.2024.11022362.
- S. Sakib, M. S. Sadi, M. S. Arefin, M. R. Islam and **H. Neha**, "An Assistive Device for the Visually Impaired for Enhanced Navigation and Reading," 2024 13th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 2024, pp. 767-772, doi: 10.1109/ICECE64886.2024.11024586.
- **H. Neha**, S. Sakib, F. Sadaf and S. M. Taslim Uddin Raju, "Mobile Application to Collect Data and Measure Blood Component Level in a Non-Invasive Way," 2023 26th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2023, pp. 1-6, doi: 10.1109/ICCIT60459.2023.10441186.

Achievements

- **Top 3 CGPA** among 120 students in the CSE department, Khulna University of Engineering & Technology (KUET).
- **Dean's List Award** (3 times) for academic excellence, KUET.
- **Top-10 Finalist**, UNDP Women's Digital Innovation Hackathon 2021, for a community app for parents of autistic children.